# Mihir Ravindra Adelkar

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Software Engineer with 3+ years building production systems at scale. Led a ground-up microservices migration across 20+ engineers, 12 services, 45% infra cost reduction. Shipped AI pipelines. Strong foundation in distributed systems, cloud infra, and full-stack product engineering. Drawn to systems where scale makes every architectural decision consequential.

## Education

**Northeastern University** — Boston, MA
*Master of Science in Software Engineering Systems* — *Sept 2023 – May 2025*

**University of Mumbai** — Mumbai, India
*Bachelor of Engineering in Computer Engineering* — *Aug 2018 – Jun 2021*

## Technical Skills

**Languages**: Python, JavaScript, Java, TypeScript, C++, SQL, Shell
**Frameworks**: React, Next.js, Node.js, Django, FastAPI, Spring Boot, GraphQL, gRPC
**AI & Data**: Claude API, RAG, Pinecone, MCP, Ollama, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch
**Cloud & Infra**: AWS, GCP, Docker, Kubernetes, Terraform, EKS, Lambda, S3, GitHub, Jenkins, Helm
**Observability & Testing**: Prometheus, New Relic, Jest, Cypress, K6

## Experience

**Full-Stack Developer** — Aug 2025 – Present
*KeelWorks Foundation* — *Boston, MA*

- Own end-to-end development of KeelCompass, an internal workforce engagement forum; deliver React UI and Node, PostgreSQL microservices partnering with PM and design, enabling async collaboration for a distributed team
- Automated end-to-end deployment pipeline with GitHub Actions on AWS EKS, eliminating manual release steps and unblocking same-day deploys for the engineering team
- Delivered reusable component library with coding standards adopted across the team, cutting repeated code patterns and reducing onboarding time for new contributors

**Software Engineer** — May 2023 – Sep. 2023
*Freelancing (Dabadu.ai)* — *Remote*

- Resolved production defects across XRM automotive dealership CRM; diagnosed stability issues across multiple modules, restoring reliable deployments for live dealership operations
- Refactored legacy JavaScript to TypeScript and introduced SonarCloud quality gates, reducing production bugs 20%

**Software Engineer** — Jun. 2021 – May 2023
*NeoSOFT Technologies* — *Mumbai, India*

- Built Open Source boilerplates - mf-shell (Webpack Module Federation enabling micro-frontend composition) and Console (multi-tenant admin panel with RBAC); Adopted as org-wide scaffolding standards by 15+ internal teams
- Audited 500K+ user real estate CRM (400–500ms avg, 2s spikes); designed full architecture, flow diagrams, and service migration plan; built pre-configured gRPC/Kafka microservices boilerplate; coordinated 20+ engineers over 6 months – 12 services migrated, 7x faster inter-service calls, 45%+ infra cost reduction via AWS rightsizing
- Engineered Homepage Builder CMS and multi-tenant CRM for Nesto using React, Node.js, and AWS Lambda; enabled marketing to independently update pages and promotions, eliminating a recurring engineering bottleneck
- Implemented CI/CD with mandatory tests and New Relic APM dual dashboards (engineering metrics + stakeholder health view), cutting MTTR from 25 to 15 minutes
- Architected Django analytics engine aggregating per-student classroom data, visualized on Next.js dashboard – enabled teachers to monitor live attendance, quiz scores, and engagement during class with post-session reports
- Designed GraphQL/REST APIs with PostgreSQL and Redis caching serving 500K+ users, cutting response times 70%; established Jest/Cypress automation reaching 85% test coverage
- Mentored 8 junior developers through code reviews and initiated bi-weekly knowledge sharing seminars

## Recent Projects

**Healthcare Prior Authorization AI** | *Python, FastAPI, Claude API, FHIR R4, MCP, Pinecone, Streamlit*

- Engineered AI system replacing 3–5 day manual insurance reviews with automated decisions in 5–10 sec; built clinical NLP pipeline mapping unstructured notes to FHIR R4 resources with 95%+ extraction accuracy on test dataset
- Designed MCP architecture for context-aware LLM reasoning combining patient data with insurance policy documents; system outputs confidence scores with full audit trails enabling reviewers to prioritize high-value claims

**CVE Intelligence System** | *Go, Node.js, Kafka, Pinecone, Llama 3, AWS EKS, Terraform, Istio*

- Built distributed microservices for real-time CVE ingestion using Kafka streaming; developed RAG pipeline with Pinecone and Llama 3 enabling engineers to query vulnerabilities in plain English instead of manual NVD searches
- Deployed on AWS EKS with Terraform IaC and Istio service mesh; implemented Prometheus/Grafana monitoring with automated alerting and Jenkins CI/CD pipeline